

Supplementary Information

Oxygen Evolution Reaction: A perspective on a decade of atomic scale simulations

Spyridon Divanis^a, Tugce Kutlusoy^a, Ida Marie Ingmer Boye^a, Isabela Costinela Man^b,
Jan Rossmeisl^a

^a Center of Excellence (CoE), Department of Chemistry, University of Copenhagen,
Universitetsparken 5, 2100 København Ø, Copenhagen, Denmark.

^b C.D. Nenitzescu, Center for Chemistry, Romanian Academy, Spl. Independetei 202b,
Bucharest, Romania.

Methods

We subtracted the data points (ΔE) from a total collection of 24 papers. We converted Gibbs free energy (ΔG) values into the ΔE when it was needed. Because we used ΔE values on the graphs related to scaling relation. ΔE is calculated by the following expression, $\Delta E = \Delta G - \Delta ZPE + T\Delta S$, where ΔZPE is the difference in Zero Point Energy due to the reaction and ΔS the change in entropy. In the case that the ΔZPE and ΔS were not provided, the corrections from Rossmeisl et al. [25] were applied to the data in order to obtain ΔE values. Furthermore, to the construction of the volcano plot in Fig.3, we used the ΔG values calculated by the following expression $\Delta G = \Delta E + \Delta ZPE - T\Delta S$, where we used the same ΔZPE and ΔS from Table SI-1 for all the data.

Table SI-1: Zero Point Energy (ZPE) and entropy corrections [25]

	TS	$T\Delta S$	ZPE	ΔZPE	$\Delta ZPE - T\Delta S$
H ₂ O	0,67	0	0,56	0	0
*OH + ½ H ₂	0,20	-0,47	0,44	-0,12	0,35
*O + H ₂	0,41	-0,27	0,34	-0,22	0,05
½ O ₂ + H ₂	0,73	0,05	0,32	-0,24	-0,29
H ₂	0,41		0,27		
½ O ₂	0,32		0,05		
O*	0		0,07		
HO*	0		0,30		
H*	0		0,17		

Table SI-2: Total data from the 24 papers

Article	Structure	ΔE^*_{OH} (eV)	ΔE^*_{O} (eV)	ΔE^*_{OOH} (eV)	ΔE_{O_2} (eV)
1	MoO2	-0.02	0.66	3.02	4.92
1	IrO2	-0.23	1.32	2.79	4.92
1	RuO2	0.60	2.34	3.47	4.92
1	TiO2	1.73	4.57	4.86	4.92
1	RhO2	0.68	2.64	3.63	4.92
1	NbO2	-0.97	-0.86	2.36	4.92
1	VO2	0.25	0.48	3.52	4.92
1	MnO2	1.09	2.70	4.15	4.92
1	CrO2	1.12	1.64	4.15	4.92
1	PtO2b	0.83	2.3	3.89	4.92
1	PbO2b	0.89	2.34	4.21	4.92
1	SnO2b	-0.05	0.52	3.41	4.92
1	SrTiO3	1.22	3.88	4.51	4.92
1	SrVO3	-0.26	-0.10	2.69	4.92
1	SrCrO3	0.37	0.89	3.60	4.92
1	SrMnO3	0.82	2.27	4.02	4.92
1	SrFeO3	1.29	2.95	4.30	4.92
1	SrCoO3	1.15	2.97	4.06	4.92
1	SrNiO3	1.79	3.82	4.54	4.92
1	SrCuO3	1.94	4.72	4.92	4.92
1	SrRuO3	0.72	2.23	3.69	4.92
1	LaTiO3	-1.55	-1.85	1.84	4.92
1	LaVO3	-0.61	-0.78	2.57	4.92
1	LaCrO3	0.17	0.66	3.30	4.92
1	LaMnO3	0.27	1.36	3.47	4.92
1	LaFeO3	0.79	2.27	3.87	4.92
1	LaCoO3	1.10	2.64	4.12	4.92
1	LaNiO3	1.15	3.07	4.22	4.92
1	LaCuO3	2.03	4.89	5.00	4.92
1	LaRuO3	0.35	1.84	3.35	4.92
1	Mn3O4	0.56	2.04	3.76	4.92
1	Mn2O3(1)	-0.19	1.25	3.13	4.92
1	Mn2O3(2)	-0.15	0.93	3.26	4.92
1	MnO2	1.14	2.75	4.24	4.92
1	Co3O4	1.02	2.74	3.94	4.92
1	NiO	0.46	2.27	3.51	4.92
1	MoO2	0.05	1.02	3.40	4.92

1	IrO2		-0.10	1.61	3.02	4.92
1	RuO2		0.96	2.68	3.92	4.92
1	RhO2		0.76	2.80	3.76	4.92
1	NbO2		-0.83	-0.51	2.64	4.92
1	ReO2		-1.10	-0.64	2.56	4.92
1	VO2		0.16	0.78	3.75	4.92
1	MnO2		1.38	3.08	4.56	4.92
1	CrO2		1.42	2.14	4.98	4.92
1	PtO2b		0.82	2.31	3.79	4.92
1	NiO2b		0.65	2.89	3.85	4.92
1	PbO2b		0.67	2.98	4.04	4.92
1	SnO2b		-1.02	0.17	2.55	4.92
1	Mn3O4		0.69	2.4	3.85	4.92
1	Mn2O3 (II)		0.84	2.29	3.92	4.92
1	Co3O4		1.02	2.74	3.94	4.92
2	TiO2 (Ti16O32)		1.68	4.47	4.52	4.92
2	V-Ti15O32	5c-M (cus(M))	0.98	2.22	3.93	4.92
2	5c-M (cus(Ti))		0.79	3.04	4.11	4.92
2	6c-M (cus(Ti))		-0.01	2.63	3.41	4.92
2	6c-Ms1 (cus(Ti))		0.63	3.27	4.00	4.92
2	6c-Ms2 (cus(Ti))		0.73	3.34	4.07	4.92
2	Nb-Ti15O32	5c-M (cus(M))	-0.48	1.26	2.83	4.92
2	5c-M (cus(Ti))		-0.16	2.01	3.18	4.92
2	6c-M (cus(Ti))		-0.29	2.21	3.05	4.92
2	6c-Ms1 (cus(Ti))		-0.34	2.26	3.03	4.92
2	6c-Ms2 (cus(Ti))		-0.10	2.57	3.24	4.92
2	Ta-Ti15O32	5c-M (cus(M))	-0.91	1.37	2.48	4.92
2	5c-M (cus(Ti))		-0.42	1.73	2.96	4.92
2	6c-M (cus(Ti))		-0.32	2.21	3.04	4.92
2	6c-Ms1 (cus(Ti))		-0.75	2.04	2.81	4.92
2	6c-Ms2 (cus(Ti))		-0.27	2.42	3.08	4.92
2	Cr-Ti15O32	5c-M (cus(M))	1.46	2.3	4.31	4.92
2	5c-M (cus(Ti))		1.22	3.32	4.58	4.92
2	6c-M (cus(Ti))		0.84	2.78	4.2	4.92
2	6c-Ms1 (cus(Ti))		1.33	4.02	4.67	4.92
2	6c-Ms2 (cus(Ti))		1.17	3.72	4.48	4.92
2	Mo-Ti15O32	5c-M (cus(M))	0.03	0.36	3.16	4.92
2	5c-M (cus(Ti))		0.22	1.34	3.6	4.92
2	6c-M (cus(Ti))		-0.11	0.73	3.27	4.92
2	6c-Ms1 (cus(Ti))		0.23	2.06	3.61	4.92
2	6c-Ms2 (cus(Ti))		0.19	2.22	3.54	4.92
2	W-Ti15O32	5c-M (cus(M))	-0.63	-0.46	2.6	4.92

2	5c-M (cus(Ti))		-0.05	0.29	3.34	4.92
2	6c-M (cus(Ti))		-0.34	0.13	2.94	4.92
2	6c-Ms1 (cus(Ti))		-0.34	1.03	3.04	4.92
2	6c-Ms2 (cus(Ti))		-0.18	1.34	3.15	4.92
2	Mn-Ti15O32	5c-M (cus(M))	1.53	3.26	4.26	4.92
2	5c-M (cus(Ti))		1.67	4.09	4.51	4.92
2	6c-M (cus(Ti))		1.55	4.07	4.53	4.92
2	6c-Ms1 (cus(Ti))		1.62	4.37	4.51	4.92
2	6c-Ms2 (cus(Ti))		1.62	4.36	4.54	4.92
2	Fe-Ti15O32	5c-M (cus(M))	1.71	3.6	4.39	4.92
2	5c-M (cus(Ti))		1.77	4.19	4.54	4.92
2	6c-M (cus(Ti))		1.16	3.68	4.42	4.92
2	6c-Ms1 (cus(Ti))		1.16	3.85	4.50	4.92
2	6c-Ms2 (cus(Ti))		1.11	3.76	4.38	4.92
2	Ru-Ti15O32	5c-M (cus(M))	0.66	2.02	3.63	4.92
2	5c-M (cus(Ti))		1.34	3.42	4.56	4.92
2	6c-M (cus(Ti))		0.61	2.85	3.97	4.92
2	6c-Ms1 (cus(Ti))		1.03	3.71	4.40	4.92
2	6c-Ms2 (cus(Ti))		1.08	3.77	4.39	4.92
2	Ir-Ti15O32	5c-M (cus(M))	0.22	1.68	3.12	4.92
2	5c-M (cus(Ti))		1.36	3.33	4.12	4.92
2	6c-M (cus(Ti))		0.74	2.6	4.08	4.92
2	6c-Ms1 (cus(Ti))		1.05	3.65	4.38	4.92
2	6c-Ms2 (cus(Ti))		0.96	3.52	4.30	4.92
2	Ni-Ti15O32	5c-M (cus(M))	1.64	4.42	4.23	4.92
2	5c-M (cus(Ti))		1.99	4.46	4.56	4.92
2	6c-M (cus(Ti))		1.64	4.14	4.75	4.92
2	6c-Ms1 (cus(Ti))		1.69	4.44	4.52	4.92
2	6c-Ms2 (cus(Ti))		1.65	4.34	4.50	4.92
3	Fe2O3(0001) - Ti (U)		-0.86	-0.24	2.56	4.92
3	Fe2O3(0001) - Mn (U)		0.01	1.4	3.42	4.92
3	Fe2O3(0001) Co (U)		0.42	2.55	3.69	4.92
3	Fe2O3(0001) Ni (U)		0.59	2.64	3.84	4.92
3	Fe2O3(0001) Si (U)		-0.36	0.54	2.99	4.92
3	F2O3(0001) F (U)		-0.52	1.95	3.04	4.92
3	Fe2O3 - Ti (U)		-0.33	0.41	3.26	4.92
3	Fe2O3 - Mn (U)		0.34	1.81	3.91	4.92
3	Fe2O3 Co (U)		0.86	3.05	4.29	4.92
3	Fe2O3 Ni (U)		1.11	3.25	4.51	4.92
3	Fe2O3 Si-H2O (U)		-0.58	0.39	2.99	4.92
3	Fe2O3 F-H2O (U)		0.02	2.77	3.58	4.92

3	Fe2O3(0001) - Ti (U)	-0.31	0.28	3.10	4.92
3	Fe2O3(0001) - Mn (U)	0.83	2.4	4.24	4.92
3	Fe2O3(0001) Co (U)	1.18	3.34	4.46	4.92
3	Fe2O3(0001) Ni (U)	1.34	3.46	4.60	4.92
3	Fe2O3(0001) - Ti (U)	-0.30	0.34	3.17	4.92
3	Fe2O3(0001) - Mn (U)	0.61	2.06	3.96	4.92
3	Fe2O3(0001) Co (U)	1.11	3.20	4.36	4.92
3	Fe2O3(0001) Ni (U)	1.30	3.36	4.53	4.92
3	Fe2O3(0001) Si (U)	-0.07	0.93	3.40	4.92
3	Fe2O3(0001) (U)	0.30	2.71	3.67	4.92
3	Fe2O3(0001) (U)	0.00	2.27	3.57	4.92
3	Fe2O3(0001)- (U)	-0.38	1.83	3.05	4.92
4	Co3O4 RBPE (001)	0.9	2.56	3.76	4.92
4	B-CoOOH RPBE (01-12)	0.81	2.62	3.68	4.92
4	Co3O4 RBPE+U (001) (U)	1.55	3.71	4.32	4.92
4	B-CoOOH RPBE+U(01-12) (U)	1.57	3.72	4.34	4.92
4	Co3O4 DACAPO- RPBE	1.02	2.74	3.94	4.92
5	ScO	-1.83	-2.52	1.579	4.92
5	TiO	-1.32	-1.66	1.929	4.92
5	VO	-0.96	-1.01	2.166	4.92
5	CrO	-0.33	0.177	2.506	4.92
5	MnO	0.27	1.372	3.283	4.92
5	FeO	0.25	1.707	3.358	4.92
5	CoO	0.68	1.946	3.519	4.92
5	NiO	0.94	2.424	3.727	4.92
5	CuO	2.01	3.882	4.352	4.92
5	(La/Y)ScO3	-1.581	-1.878	1.826	4.92
5	(La/Y)TiO3	-1.39	-1.663	2.072	4.92
5	(La/Y)VO3	-0.61	-0.81	2.555	4.92
5	(La/Y)CrO3	0.01	0.488	3.293	4.92
5	(La/Y)MnO3	0.248	1.301	3.464	4.92
5	(La/Y)FeO3	0.77	2.23	3.87	4.92
5	(La/Y)CoO3	1.08	2.6	4.097	4.92
5	(La/Y)NiO3	1.147	3.07	4.22	4.92
5	(La/Y)CuO3	2.023	4.85	4.99	4.92
5	(Sr/Ca/Ba)VO3	-0.271	-0.11	2.953	4.92
5	(Sr/Ca/Ba)CrO3	0.346	0.87	3.607	4.92
5	(Sr/Ca/Ba)MnO3	0.757	2.233	4.042	4.92
5	(Sr/Ca/Ba)FeO3	1.266	2.95	4.307	4.92
5	(Sr/Ca/Ba)CoO3	1.125	2.97	4.051	4.92
5	(Sr/Ca/Ba)NiO3	1.45	3.7	4.458	4.92

5	(Sr/Ca/Ba)CuO ₃	1.926	4.69	4.751	4.92
5	NiO	0.48	2.29	3.57	4.92
5	LaNiO ₃	1.15	3.06	4.21	4.92
5	YNiO ₃	1.25	3.25	4.31	4.92
5	BaNiO ₃	1.46	3.71	4.44	4.92
5	SrNiO ₃	1.77	3.80	4.52	4.92
5	Mn ₂ O ₃ (110) 1	-0.21	1.24	3.11	4.92
5	Mn ₂ O ₃ (110) 2	0.24	1.69	3.44	4.92
5	Mn ₂ O ₃ (110) 3	0.40	1.73	3.75	4.92
5	Mn ₂ O ₃ (110) 4	0.59	2.09	3.74	4.92
6	CoOOH (01-12) (U)	1.59	3.77	4.35	4.92
6	CoOOH (01-14) (U)	0.47	2.31	3.58	4.92
6	CoOOH (0001) (U)	-0.15	1.70	3.29	4.92
6	CoOOH (01-14) + Mn (U)	0.10	1.51	3.35	4.92
6	CoOOH (01-14) + Fe (U)	0.05	2.00	3.20	4.92
6	CoOOH (01-14) + Co (U)	0.47	2.31	3.58	4.92
6	CoOOH (01-14) + Ni (U)	0.80	2.65	3.78	4.92
6	CoOOH (01-12) + V (U)	0.85	2.75	4.10	4.92
6	CoOOH (01-14) + Cu (U)	1.30	3.50	4.20	4.92
6	CoOOH (01-12) +Co (U)	1.59	3.77	4.35	4.92
6	CoOOH (01-12) + Mn (U)	1.45	3.90	4.50	4.92
7	SnO ₂	1.65	5.22	4.92	4.92
7	SnO ₂ red	-0.50	0.45	3.02	4.92
7	PtO ₂	0.46	2.89	3.56	4.92
7	PbO ₂	2.27	5.71	5.22	4.92
7	SrTiO ₃	0.77	3.46	4.15	4.92
7	BaTiO ₃	-0.31	1.53	3.11	4.92
7	CdO	2.14	5.38	4.82	4.92
7	MgO	2.56	5.84	5.06	4.92
7	TiO ₂	1.611	4.40	4.86	4.92
7	NaTaO ₃	-1.09	-0.55	2.86	4.92
7	SnO ₂	1.62	4.44	4.39	4.92
7	SnO ₂ red	-0.42	0.48	3.20	4.92
7	PtO ₂	0.26	2.53	3.48	4.92
7	PbO ₂	2.14	4.71	4.73	4.92
7	SrTiO ₃	0.92	3.61	3.84	4.92
7	BaTiO ₃	-0.38	1.43	3.07	4.92
7	CdO	2.10	4.61	4.49	4.92
7	MgO	2.36	4.55	4.72	4.92
7	TiO ₂	1.65	4.39	4.36	4.92
7	NaTaO ₃	-0.73	-0.65	2.98	4.92

7	SnO2	1.55	4.33	3.74	4.92
7	SnO2 red	-0.3	0.83	3.39	4.92
7	PtO2	0.23	2.39	3.40	4.92
7	PbO2	2.03	4.76	4.69	4.92
8	RuO2-Ni	0.83	2.24	3.81	4.92
8	RuO2-Co	0.68	2.14	3.60	4.92
8	RuO2 DACAPO	0.96	2.68	3.92	4.92
8	RuO2 GPAW	0.84	2.29	3.65	4.92
8	RuO2-Ni DACAPO	0.96	2.70	3.99	4.92
9	Co3O4	1.02	2.74	3.94	4.92
9	B-CoOOH (0112) (U)	1.59	3.77	4.35	4.92
9	B-CoOOH (0114) (U)	0.47	2.31	3.58	4.92
9	B-CoOOH (0001) (U)	-0.15	1.70	3.29	4.92
9	MnO2	1.24	3.04	4.42	4.92
9	Mn2O3	0.51	2.03	3.9	4.92
10	2Ni(bridge) γ -NiOOH (U)	0.73	2.62	3.98	4.92
10	2Ni(bridge) γ -NiOOH:Fe (U)	0.77	2.92	4.03	4.92
10	Ni(on-top) γ -NiOOH (U)	1.40	3.92	4.34	4.92
10	2 Fe(bridge) γ -FeOOH (U)	-0.28	1.70	3.03	4.92
10	2Fe(bridge) γ -FeOOH:Ni (U)	0.41	2.38	3.61	4.92
10	Fe (on-top) γ -NiOOH (U)	0.68	2.60	3.84	4.92
11	(101) surface of Na2CoP2O7 (U)	0.34	1.89	2.44	4.92
11	tetrahedral oxygen vacancy formation (U)	-0.29	0.83	2.7	4.92
11	Absorbed oxygenbipyramidal formation (U)	1.57	2.70	3.28	4.92
11	(010) surface of Na-CoPO4 (U)	0.75	2.46	2.14	4.92
11	β -CoOOH (U) (1014)	0.17	1.49	2.37	4.92
12	β -CoOOH (01-12) (U)	0.75	3.27	3.59	4.92
12	γ -FeOOH (010) (U)	-0.63	1.80	2.51	4.92
12	WO3 (001) (U)	1.95	4.45	4.32	4.92
12	Fesub: β -CoOOH (01-12) (U)	0.45	2.54	3.88	4.92
12	Wtop: β -CoOOH (01-12) (U)	1.21	3.50	4.08	4.92
12	FesubWtop: β -CoOOH (01-12) (U)	0.87	3.13	3.93	4.92
12	CoWO4 (010) (U)	1.20	3.69	4.23	4.92
13	Co3O4 (001)	0.36	2.59	2.90	4.92
14	CoO3 bridge (U)	1.55	3.65	4.20	4.92
14	FexCo3-xO4 (X 25%) bridge (U)	1.48	3.50	4.28	4.92
14	FexCo3-xO4 (X 50%) bridge (U)	1.56	3.78	4.40	4.92
14	NixCo3-xO4 (X 25%) bridge (U)	1.32	3.23	4.02	4.92
14	NixCo3-xO4 (X 50%) bridge (U)	1.40	3.30	4.13	4.92
14	CuxCo3-xO4 (X 25%) hollow (U)	1.20	3.05	3.96	4.92
14	CuxCo3-xO4 (X 50%) hollow (U)	1.35	3.20	4.07	4.92

15	Bilayers: top of the metal-edge (U)	1.00	3.00	4.10	4.92
15	Trilayers: bottom of O-edge near Au (U)	-0.20	1.90	2.90	4.92
15	Trilayers: top of the metal-edge (U)	1.10	3.00	4.20	4.92
16	IrO ₂ O-coverage	-0.01	1.55	3.10	4.92
16	IrO ₂ O-coverage with explicit solvent	-0.17	1.54	2.72	4.92
17	<i>B</i> -NiOOH (001)-half O-half OH (U)	0.57	2.22	3.8	4.92
17	β -NiOOH(001)-hydroxylated completely (U)	0.75	2.33	3.67	4.92
18	CoOOH nanosheets pristine (U)	-0.22	1.10	3.00	4.92
18	CoOOH nanosheets V (U)	0.63	2.15	3.96	4.92
18	CoOOH nanosheets Cr (U)	-0.33	1.76	3.01	4.92
18	CoOOH nanosheets Mn (U)	0.52	1.48	3.88	4.92
18	CoOOH nanosheets Fe (U)	-0.27	1.51	3.09	4.92
18	CoOOH nanosheets Ni (U)	-0.11	1.21	3.02	4.92
18	CoOOH nanosheets Cu (U)	0.18	2.16	3.49	4.92
18	CoOOH nanosheets Ru (U)	-1.11	0.9	2.76	4.92
18	CoOOH nanosheets Rh (U)	-0.08	1.15	3.15	4.92
18	CoOOH nanosheets Pd (U)	0.06	1.31	3.32	4.92
18	CoOOH nanosheets Ir (U)	0.11	1.15	3.36	4.92
18	CoOOH nanosheets Pt (U)	-0.60	0.58	2.69	4.92
18	CoOOH nanosheets Ag (U)	0.45	2.26	3.89	4.92
18	CoOOH nanosheets Al (U)	0.04	2.24	3.34	4.92
18	CoOOH nanosheets Ga (U)	0.03	2.24	3.32	4.92
18	CoOOH nanosheets In (U)	0.08	2.10	3.46	4.92
18	CoOOH nanosheets Sn (U)	-0.59	0.83	2.83	4.92
18	CoOOH nanosheets Pb (U)	-0.05	1.51	2.86	4.92
18	CoOOH nanosheets Bi (U)	0.2	2.18	3.59	4.92
18	CoOOH nanosheets Mg (U)	0.29	2.74	3.64	4.92
18	CoOOH nanosheets Sc (U)	-0.32	1.84	2.92	4.92
18	CoOOH nanosheets Y (U)	0.78	2.93	4.29	4.92
18	CoOOH nanosheets Ti (U)	0.70	2.25	3.98	4.92
18	CoOOH nanosheets Nb (U)	0.36	2.27	3.69	4.92
18	CoOOH nanosheets Zn (U)	0.31	2.3	3.57	4.92
18	CoOOH nanosheets Cd (U)	0.46	2.38	3.58	4.92
18	NiOOH nanosheets pristine (U)	0.5	1.66	3.14	4.92
18	NiOOH nanosheets V (U)	-0.34	2.17	3.46	4.92
18	NiOOH nanosheets Cr (U)	0.70	1.75	2.83	4.92
18	NiOOH nanosheets Mn (U)	0.05	2.29	4.36	4.92
18	NiOOH nanosheets Fe (U)	0.38	1.89	3.2	4.92
18	NiOOH nanosheets Co (U)	0.21	1.71	3.45	4.92
18	NiOOH nanosheets Cu (U)	0.44	2.31	3.3	4.92
18	NiOOH nanosheets (U)	-0.06	2.34	3.57	4.92

18	NiOOH nanosheets Rh (U)	-0.19	1.46	2.91	4.92
18	NiOOH nanosheets Pd (U)	-0.31	1.26	2.83	4.92
18	NiOOH nanosheets Ir (U)	-0.38	1.39	2.82	4.92
18	NiOOH nanosheets Pt (U)	0.54	1.05	2.71	4.92
18	NiOOH nanosheets Ag (U)	0.11	2.07	3.47	4.92
18	NiOOH nanosheets Al (U)	-0.01	2.25	3.21	4.92
18	NiOOH nanosheets Ga (U)	-0.09	2.16	3.10	4.92
18	NiOOH nanosheets In (U)	-0.04	1.96	3.13	4.92
18	NiOOH nanosheets Sn (U)	0.15	1.46	3.09	4.92
18	NiOOH nanosheets Pb (U)	0.10	1.66	3.23	4.92
18	NiOOH nanosheets Bi (U)	0.06	1.56	3.19	4.92
18	NiOOH nanosheets Mg (U)	-0.25	2.32	3.22	4.92
18	NiOOH nanosheets Sc (U)	-0.39	2.02	2.99	4.92
18	NiOOH nanosheets Y (U)	0.47	1.78	3.00	4.92
18	NiOOH nanosheets Ti (U)	0.43	2.34	3.43	4.92
18	NiOOH nanosheets Nb (U)	0.13	2.59	3.62	4.92
18	NiOOH nanosheets Zn (U)	0.19	2.16	3.21	4.92
18	NiOOH nanosheets Cd (U)	-0.35	2.11	3.36	4.92
18	CoOOH nanosheet edge pristine (U)	0.54	2.78	3.61	4.92
18	CoOOH nanosheet edge Cr (U)	0.50	2.80	3.64	4.92
18	CoOOH nanosheet edge Fe (U)	0.73	2.96	3.8	4.92
18	NiOOH bulk terrace pristine (U)	0.66	2.69	3.60	4.92
18	NiOOH bulk terrace V (U)	-0.11	1.77	3.03	4.92
18	NiOOH bulk terrace Cr (U)	-0.03	1.99	2.97	4.92
18	NiOOH bulk terrace Fe (U)	0.08	1.91	3.29	4.92
18	NiOOH bulk terrace Ru (U)	-0.21	1.35	2.93	4.92
18	NiOOH bulk terrace Rh (U)	-0.09	1.61	2.98	4.92
18	NiOOH bulk terrace Ir (U)	-0.66	0.99	2.48	4.92
18	NiOOH bulk edge pristine (U)	1.70	4.39	4.02	4.92
18	NiOOH bulk edge V (U)	0.69	2.88	3.74	4.92
18	NiOOH bulk edge Cr (U)	1.53	3.92	3.62	4.92
18	NiOOH bulk edge Ru (U)	0.56	1.60	3.18	4.92
18	NiOOH bulk edge Rh (U)	0.73	2.56	3.19	4.92
18	NiOOH bulk edge Ir (U)	0.18	1.71	2.84	4.92
18	NiOOH nanosheet edge pristine (U)	0.89	2.97	3.71	4.92
18	NiOOH nanosheet edge V (U)	0.24	2.68	3.08	4.92
18	NiOOH nanosheet edge Cr (U)	0.53	2.55	3.44	4.92
18	NiOOH nanosheet edge Fe (U)	0.82	3.39	3.69	4.92
18	NiOOH nanosheet edge Ru (U)	-0.45	0.45	2.36	4.92
18	NiOOH nanosheet edge Rh (U)	0.35	2.08	3.21	4.92
18	NiOOH nanosheet edge Ir (U)	-0.77	0.95	2.93	4.92

19	MnO2 Site 0 (U)	-0.67	0.33	2.71	4.92
19	MnO2 Site 1 (U)	0.99	3.10	3.57	4.92
19	MnO2 Site 2 (U)	0.58	3.18	3.81	4.92
19	MnO2 Site 3 (U)	-0.26	1.77	2.68	4.92
19	MnO2 Site 4 (U)	-0.14	1.24	2.99	4.92
19	MnO2 Site 5 (U)	-0.83	0.66	2.33	4.92
19	MnO2 Site 6 (U)	0.05	0.54	3.03	4.92
19	MnO2-Co-host-cus (U)	1.25	3.39	3.92	4.92
19	MnO2-Co-dopant-cus (U)	1.49	3.41	3.94	4.92
19	MnO2-Fe-host-cus (U)	1.79	4.02	3.8	4.92
19	MnO2-Fe-dopant-cus (U)	1.57	3.74	3.63	4.92
19	MnO2-Ni-host-cus (U)	1.53	3.56	3.72	4.92
19	MnO2-Ni-dopant-cus (U)	1.53	3.72	3.72	4.92
19	MnO2-Pd-host-cus (U)	1.17	3.14	3.6	4.92
19	MnO2-Ti-host-cus (U)	0.89	3.12	3.54	4.92
19	MnO2-Al-host-cus (U)	0.47	2.61	2.85	4.92
19	MnO2-Ga-host-cus (U)	0.61	2.77	3.01	4.92
19	MnO2-Mg-host-cus (U)	0.35	2.22	2.57	4.92
19	MnO2-Nb-host-cus (U)	0.83	3.33	3.98	4.92
19	MnO2-Cr-bridge (U)	-0.69	1.56	2.36	4.92
19	MnO2-Fe-bridge (U)	-0.44	1.92	2.77	4.92
19	MnO2-Co-bridge (U)	0.17	2.3	3.07	4.92
19	MnO2-Ni-bridge (U)	0.15	2.26	3.09	4.92
19	MnO2-Pd-bridge (U)	-0.48	1.43	2.43	4.92
19	MnO2-Pt-bridge (U)	-0.66	1.07	2.26	4.92
19	MnO2-Ti-bridge (U)	-0.26	2.11	2.99	4.92
19	MnO2-Al-bridge (U)	-0.45	2.17	2.75	4.92
19	MnO2-Ga-bridge (U)	-0.43	2.2	2.81	4.92
19	MnO2-Sc-bridge (U)	-0.68	1.64	2.6	4.92
19	MnO2-Nb-bridge (U)	-0.42	0.9	2.98	4.92
20	LaCrO3	0.02	0.8	2.96	4.92
20	LaCrO3 (U)	0.64	2.02	3.55	4.92
20	LaCrO3 (U)	0.98	2.82	3.91	4.92
20	LaMnO3	0.29	3.2	3.23	4.92
20	LaMnO3 (U)	0.91	3.22	3.13	4.92
20	LaMnO3 (U)	1.26	3.23	4.09	4.92
20	LaFeO3	0.19	3.11	3.85	4.92
20	LaFeO3 (U)	0.99	3.09	3.86	4.92
20	LaFeO3 (U)	1.45	3.14	3.82	4.92
20	LaCrO3 a 0.15	0.69	2.05	3.98	4.92
20	LaCrO3 a 0.25	0.99	2.74	4.35	4.92

20	LaCrO3 a 0.35	1.43	3.53	4.76	4.92
20	LaMnO3 a 0.15	1.01	3.32	3.52	4.92
20	LaMnO3 a 0.25	1.46	3.23	3.43	4.92
20	LaMnO3 a 0.35	1.56	3.3	3.35	4.92
20	LaFeO3 a 0.15	1.16	3.33	4.29	4.92
20	LaFeO3 a 0.25	1.7	3.44	4.42	4.92
20	LaFeO3 a 0.35	2.14	3.53	4.51	4.92
21	CoOOH (01-12) (U)	1.59	3.42	4.35	4.92
21	CoOOH (01-12) (Vo:oxygen vacancy)(U)	0.92	2.71	4.06	4.92
21	V-CoOOH (01-12) (U)	0.68	2.26	3.85	4.92
21	Mn-CoOOH (01-12) (U)	1.28	3.55	4.36	4.92
21	Ni-CoOOH (01-12) (U)	0.38	1.77	3.55	4.92
22	YRhO3	0.49	2.24	3.37	4.92
22	BaRuO3	0.19	1.9	3.06	4.92
22	BaRhO3	0.35	2.13	3.12	4.92
22	ScIrO3	0.15	1.89	3.13	4.92
22	KRuO3	0.92	2.8	3.76	4.92
22	BaIrO3	0.53	2.06	3.49	4.92
22	SrIrO3	0.36	1.85	3.33	4.92
22	BaPtO3	0.81	2.88	3.59	4.92
22	BeIrO3	1.34	3.02	4.55	4.92
22	SrRhO3	-0.35	1.87	2.53	4.92
22	LaCoO3	0.09	2.37	3.11	4.92
22	SrSbO3	-1.00	1.20	2.48	4.92
22	YNiO3	0.96	3.29	3.22	4.92
22	SrPdO3	0.99	3.34	3.42	4.92
22	LaNiO3	1.08	3.44	4.07	4.92
22	BaPdO3	1.08	3.48	3.75	4.92
22	NaNbO3	-0.91	0.65	2.50	4.92
22	PbTiO3	0.88	3.36	4.04	4.92
22	SrBiO3	0.34	2.87	3.16	4.92
22	LaRhO3	-0.70	1.76	2.24	4.92
22	AlBO3	1.99	4.50	4.87	4.92
22	BaSbO3	-1.30	1.25	2.21	4.92
22	KNbO3	-0.96	0.44	2.48	4.92
22	LaPtO3	0.14	2.78	2.99	4.92
22	CaTiO3	1.13	3.78	4.3	4.92
22	BaAuO3	1.70	4.36	4.43	4.92
22	BaBiO3	0.38	3.04	3.48	4.92
22	SrTiO3	0.96	3.62	4.09	4.92
22	LaPdO3	0.65	3.36	3.38	4.92

22	CaCuO3	1.72	4.45	4.61	4.92
22	SrAuO3	1.56	4.31	4.33	4.92
22	NaTiO3	2.16	4.93	4.74	4.92
22	KOsO3	-0.46	0.43	2.62	4.92
22	BaCuO3	1.75	4.54	4.59	4.92
22	KTiO3	1.99	4.77	4.57	4.92
22	KTaO3	-1.46	-0.01	1.98	4.92
22	LiSiO3	1.85	4.64	4.71	4.92
22	AgNbO3	-0.9	0.33	2.53	4.92
22	LaCuO3	1.74	4.53	4.36	4.92
22	BaTiO3	0.67	3.52	3.81	4.92
22	BaHgO3	1.87	4.72	4.89	4.92
22	NaSbO3	0.28	3.13	3.72	4.92
22	NaOsO3	-0.69	0.11	2.39	4.92
22	BaAgO3	1.95	4.82	4.83	4.92
22	SrHgO3	1.82	4.70	4.87	4.92
22	KSbO3	0.55	3.43	3.99	4.92
22	SrAgO3	1.84	4.72	4.44	4.92
22	KBiO3	1.54	4.45	4.65	4.92
22	NaTaO3	-1.55	-0.10	1.86	4.92
22	SrPbO3	1.23	4.15	4.34	4.92
22	MgYO3	2.34	5.28	4.79	4.92
22	BaZnO3	2.07	5.03	4.74	4.92
22	BaTlO3	1.74	4.71	4.62	4.92
22	SrTlO3	1.72	4.69	4.66	4.92
22	SrZnO3	2.1	5.07	4.75	4.92
22	MgScO3	2.42	5.30	4.83	4.92
22	BaPbO3	1.42	4.40	4.49	4.92
22	SrAlO3	1.84	4.84	4.43	4.92
22	KPbO3	1.83	4.83	4.72	4.92
22	SrGeO3	0.8	3.81	4.13	4.92
22	CaAlO3	1.93	4.94	4.52	4.92
22	LiTiO3	2.46	5.26	4.76	4.92
22	CaGeO3	0.78	3.81	3.91	4.92
22	LaZnO3	2.06	5.10	4.70	4.92
22	KSnO3	1.48	4.51	4.42	4.92
22	LiSnO3	2.01	5.04	4.60	4.92
22	BaCdO3	2.13	5.16	4.80	4.92
22	BaGaO3	1.84	4.91	4.55	4.92
22	SrCdO3	1.95	5.03	4.74	4.92
22	YCuO3	1.41	4.49	4.35	4.92

22	CaSiO3	0.30	3.38	3.69	4.92
22	SrGaO3	1.90	4.98	4.58	4.92
22	BaSnO3	1.04	4.13	4.19	4.92
22	KZrO3	1.91	5.00	4.62	4.92
22	LiZrO3	2.30	5.39	4.67	4.92
22	SrSnO3	0.91	4.00	4.09	4.92
22	CaAuO3	1.14	4.23	3.93	4.92
22	KHfO3	1.84	4.94	4.56	4.92
22	LiHfO3	2.19	5.29	4.66	4.92
22	SrZrO3	0.99	4.10	4.14	4.92
22	CaGaO3	1.89	5.00	4.60	4.92
22	BaZrO3	1.05	4.17	4.17	4.92
22	NaSnO3	1.47	4.60	4.48	4.92
22	NaZrO3	1.88	5.01	4.63	4.92
22	MgAlO3	1.73	4.89	4.51	4.92
22	NaHfO3	1.82	4.97	4.58	4.92
22	CaAgO3	1.48	4.63	4.14	4.92
22	BaInO3	1.72	4.88	4.51	4.92
22	BaScO3	1.94	5.12	4.57	4.92
22	LaGaO3	1.67	4.85	4.52	4.92
22	YAlO3	1.62	4.79	4.39	4.92
22	LaAlO3	1.69	4.87	4.4	4.92
22	BaHfO3	1.01	4.19	4.12	4.92
22	SrHfO3	0.93	4.12	4.07	4.92
22	SrInO3	1.57	4.76	4.43	4.92
22	SrScO3	1.87	5.08	4.53	4.92
22	CaYO3	1.19	4.40	4.15	4.92
22	CaScO3	1.69	4.90	4.44	4.92
22	SrYO3	1.27	4.49	4.17	4.92
22	BaYO3	1.49	4.72	4.35	4.92
22	LaScO3	1.34	4.58	4.04	4.92
22	LaMgO3	1.88	5.19	4.46	4.92
22	KMoO3	-0.49	-0.19	2.65	4.92
22	NaMoO3	-0.59	-0.37	2.56	4.92
22	YZnO3	1.43	4.97	4.14	4.92
22	ScAlO3	0.39	4.01	3.40	4.92
22	BaMoO3	-0.85	-0.78	2.26	4.92
22	InMoO3	-0.56	-0.52	2.55	4.92
22	SrVO3	-0.50	-0.38	2.70	4.92
22	SrMoO3	-0.97	-0.94	2.15	4.92
22	AgMoO3	-0.58	-0.54	2.55	4.92

22	BaVO ₃	-0.89	-0.85	2.36	4.92
22	ScSiO ₃	-0.86	-0.47	2.81	4.92
22	InNbO ₃	-1.42	-1.35	1.99	4.92
22	KWO ₃	-1.51	-1.69	1.72	4.92
22	InTaO ₃	-1.66	-1.64	1.77	4.92
22	NaWO ₃	-1.64	-1.91	1.60	4.92
22	LaTiO ₃	-1.64	-1.92	1.66	4.92
22	AgWO ₃	-1.53	-1.92	1.71	4.92
22	BaNbO ₃	-1.59	-1.97	1.69	4.92
22	SrNbO ₃	-1.74	-2.19	1.52	4.92
22	LaHfO ₃	-3.15	-3.02	0.43	4.92
23	NiFe LDH Ni site (U)	-0.47	2.03	2.82	4.92
23	NiFe LDH Fe site (U)	-0.57	1.23	2.57	4.92
23	Co 2+ -NiFe LDH Ni site (U)	0.77	3.35	3.93	4.92
23	Co 2+ -NiFe LDH Fe site (U)	0.60	2.29	3.54	4.92
23	Co 2+ -NiFe LDH Co site (U)	0.54	3.24	3.61	4.92
23	Co 3+ -NiFe LDH Ni site (U)	0.84	3.32	4.03	4.92
23	Co 3+ -NiFe LDH Fe site (U)	-0.01	2.05	3.47	4.92
23	Co 3+ -NiFe LDH Co site (U)	0.69	2.68	3.64	4.92
24	WO ₃ (200) undoped, reaction on W	2.34	5.06	5.52	4.92
24	Cr - WO ₃ (200), reaction on W	1.68	4.32	4.90	4.92
24	Mo - WO ₃ (200), reaction on W	1.80	4.34	5.25	4.92
24	Cr -WO ₃ (200), reaction on doped atom	2.51	4.67	5.20	4.92
24	Mo - WO ₃ (200), reaction on doped atom	2.01	4.24	5.35	4.92
24	Cr-WO ₃ (200), doped second layer	2.14	4.61	5.44	4.92
24	Mo-WO ₃ (200), doped second layer	2.05	4.46	5.44	4.92

Some extra graphs were created from the data list, and are presented below. **a)** The doped structures without U-correction is less steep from the dash blue trend line that bears a slope of two. **b)** In the diagram with the undoped structures without U-correction the O^* trend line (blue solid) has a slope very close to 2, because both electrons consisting the O^* bond on the surface, are provided from the valence band for the case of the semi-conductors. **c)** In the case of the undoped structures with U-correction the different values for the U, creates scattered data resulting a smearing effect for the slope of oxygen trend line. The values for all the graphs including the ones from the main text are presented in Table SI-3.

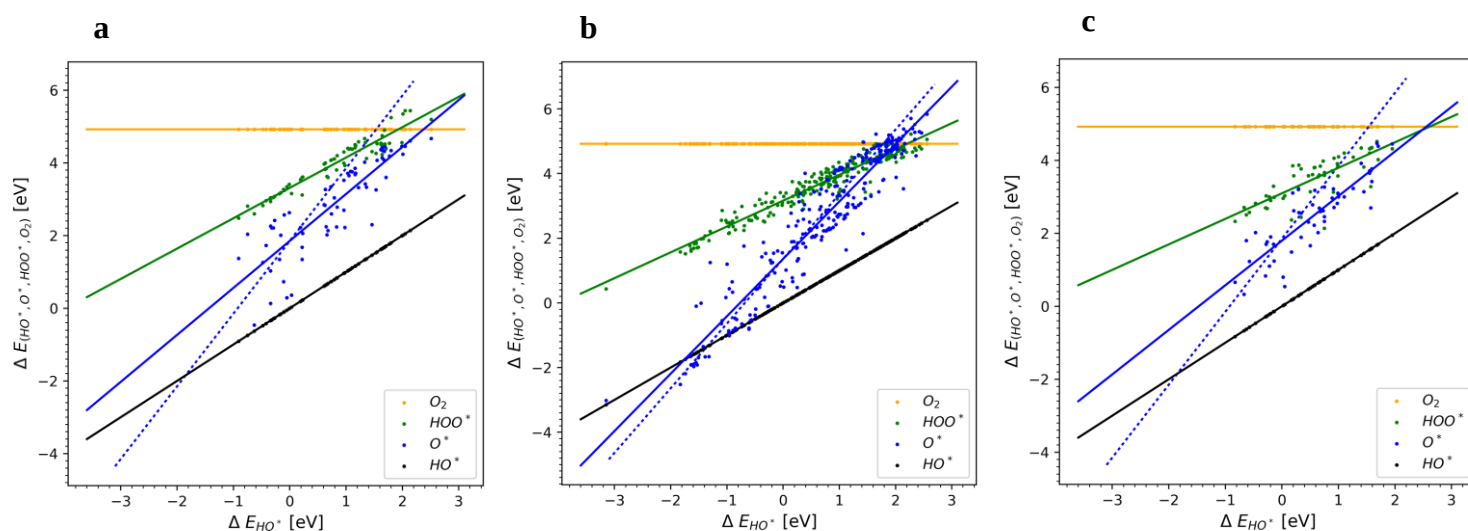


Fig SI-1: **a.** Doped (without U-correction) **b.** Undoped (without U-correction) **c.** Undoped (with U-correction)

Table SI-3: Table for the slopes of O^* trend lines

GRAPH	SLOPE of O^*
Total	1,60
Doped	1,24
Undoped	1,72
Doped + U	1,17
Doped (without U-correction)	1,29
Heavy Doped (more than 1-dopant)	1,75
TiO ₂ 2-dopant	1.90
TiO ₂ 1-dopant	1,29
Undoped + U	1,22
Undoped (without U-correction)	1,77

Density of States for the doped TiO₂ structures

4+ 5+ 6+								
22 Ti Titanium Transition Metal	23 V Vanadium Transition Metal	24 Cr Chromium Transition Metal	25 Mn Manganese Transition Metal	26 Fe Iron Transition Metal	27 Co Cobalt Transition Metal	28 Ni Nickel Transition Metal	29 Cu Copper Transition Metal	30 Zn Zinc Transition Metal
40 Zr Zirconium Transition Metal	41 Nb Niobium Transition Metal	42 Mo Molybdenum Transition Metal	43 Tc Technetium Transition Metal	44 Ru Ruthenium Transition Metal	45 Rh Rhodium Transition Metal	46 Pd Palladium Transition Metal	47 Ag Silver Transition Metal	48 Cd Cadmium Transition Metal
72 Hf Hafnium Transition Metal	73 Ta Tantalum Transition Metal	74 W Tungsten Transition Metal	75 Re Rhenium Transition Metal	76 Os Osmium Transition Metal	77 Ir Iridium Transition Metal	78 Pt Platinum Transition Metal	79 Au Gold Transition Metal	80 Hg Mercury Transition Metal
104 Rf Rutherfordium Transition Metal	105 Db Dubnium Transition Metal	106 Sg Seaborgium Transition Metal	107 Bh Bohrium Transition Metal	108 Hs Hassium Transition Metal	109 Mt Meitnerium Transition Metal	110 Ds Darmstadtium Transition Metal	111 Rg Roentgenium Transition Metal	112 Cn Copernicium Transition Metal

Fig SI-2: A screening using transition metals from the d-block as dopants in the TiO₂ structure was made to draw graph corresponding to a scaling relation as seen on Figure 8 in the main text. Both n-type and p-type were used in a total number of 23 dopants (Cr, Mn, Cu, Ru, Rh, Pd, Pt, Os, Ir, Au, V, Nb, Mo, W, Fe, Co, Ni, Zn, Cd, Ag, Re, Zr, Hf). As it was expected using only one dopant in the unit cell creates a trend line for O*, with a slope very close to 1, on the contrary using two dopants the slope is close to 2. This is happening due to the change of the binding energy of the O* for the case of the n-type dopants when two dopants are used. On the other hand, using one or two the p-type dopants in the unit cell does not affect the binding energy of the O* since even one p-type dopant is providing a hole to the system and places the Fermi level in the valence band. The density of states for all the tested dopants are provided in the following diagrams. Chosen dopants have been highlighted on periodic table (blue square frame represents host oxide (TiO₂); red square frame represents n-type dopants; green square frame represents rest of the dopants (oxidation state of dopant equal to the host (+4) or less than host (+4>)).

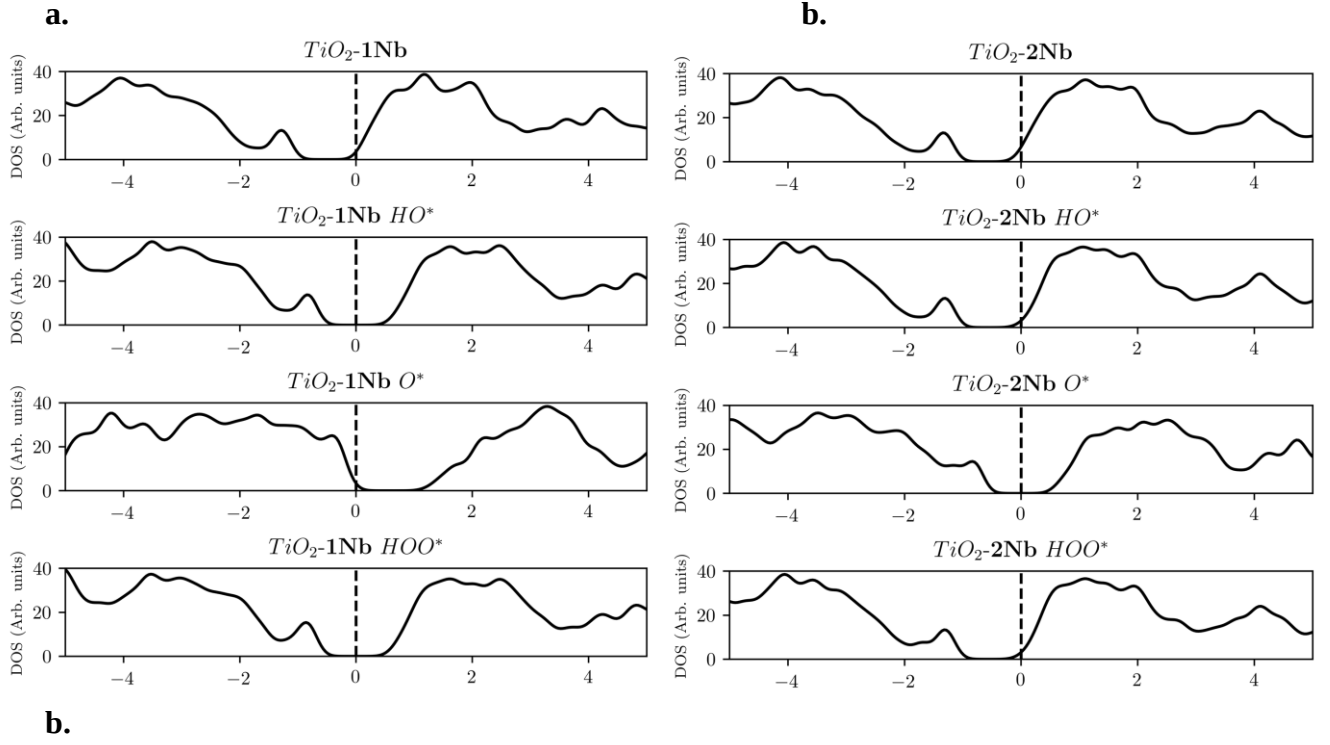


Fig SI-3: Density of state (DOS) **a.** one dopant (Nb) in TiO₂ **b.** two dopant (Nb) in TiO₂

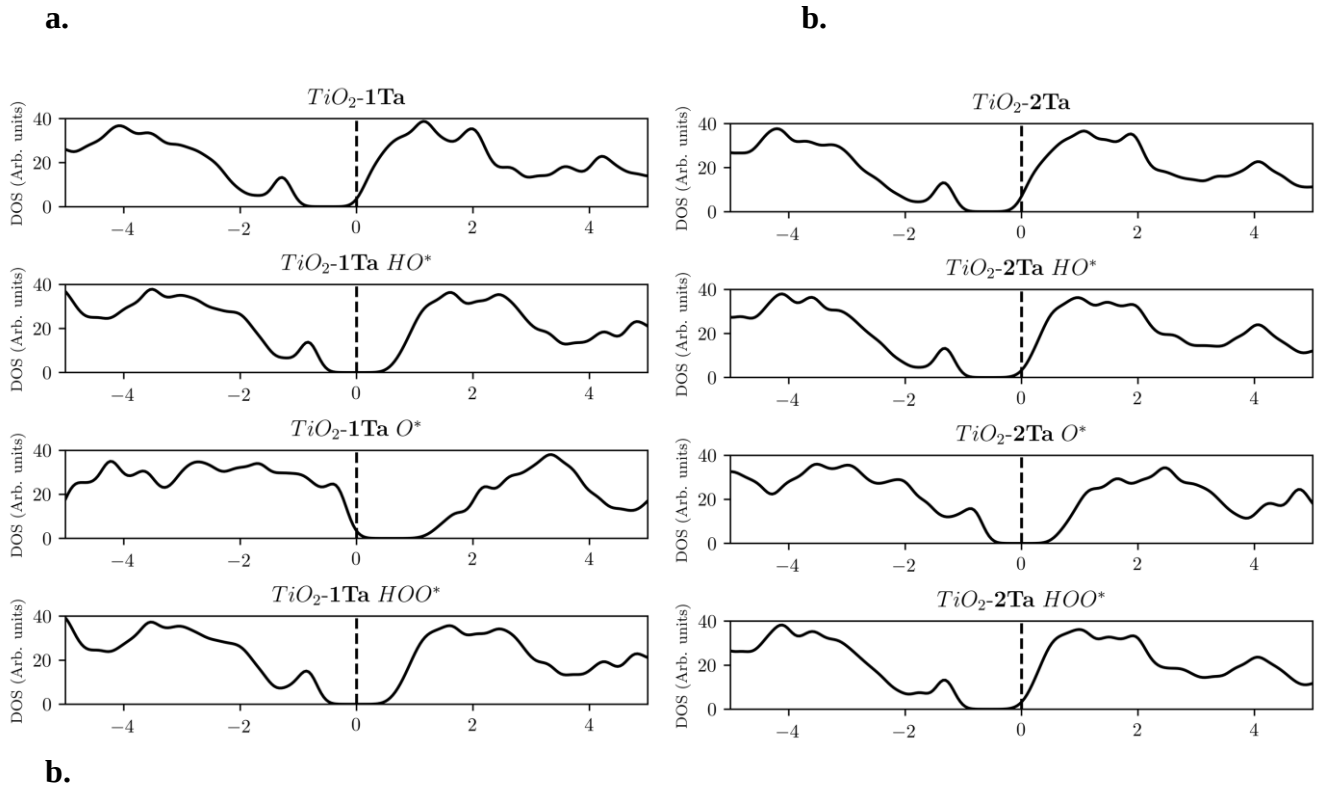


Fig SI-4: Density of state (DOS) **a.** one dopant (Ta) in TiO₂ **b.** two dopant (Ta) in TiO₂

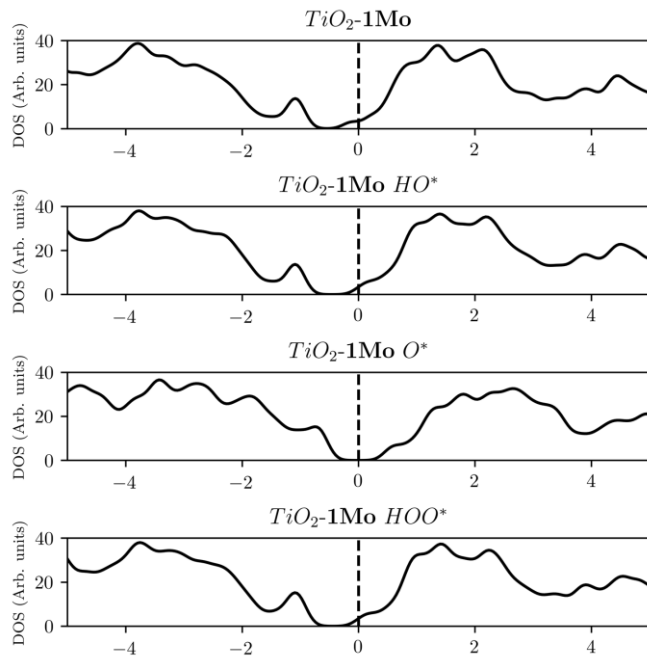
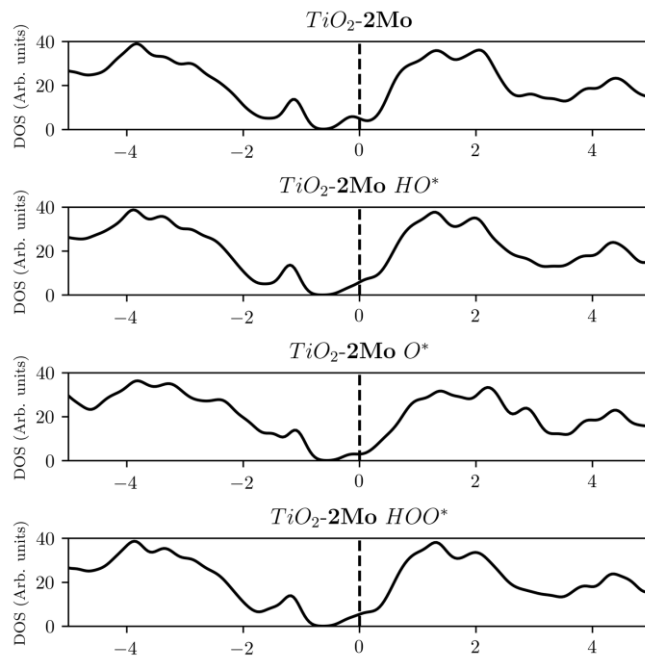
a.**b.**

Fig SI-5 Density of state (DOS) **a.** one dopant (Mo) in TiO_2 **b.** two dopant (Mo) in TiO_2

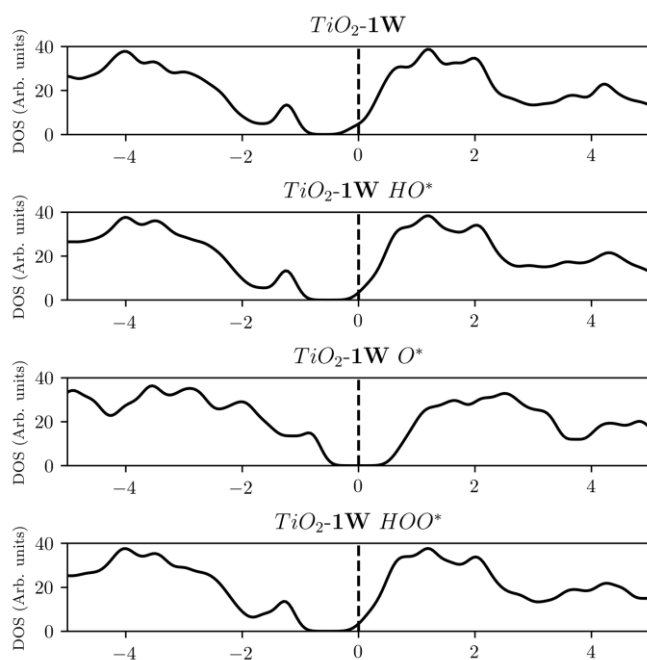
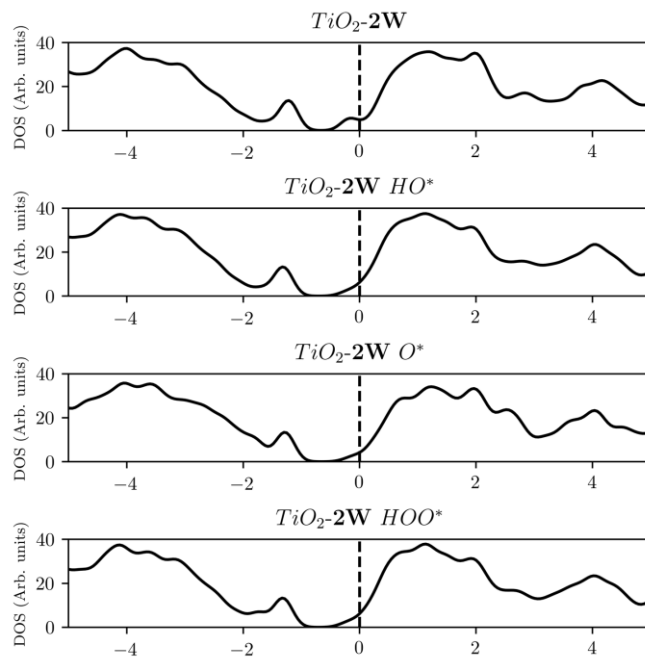
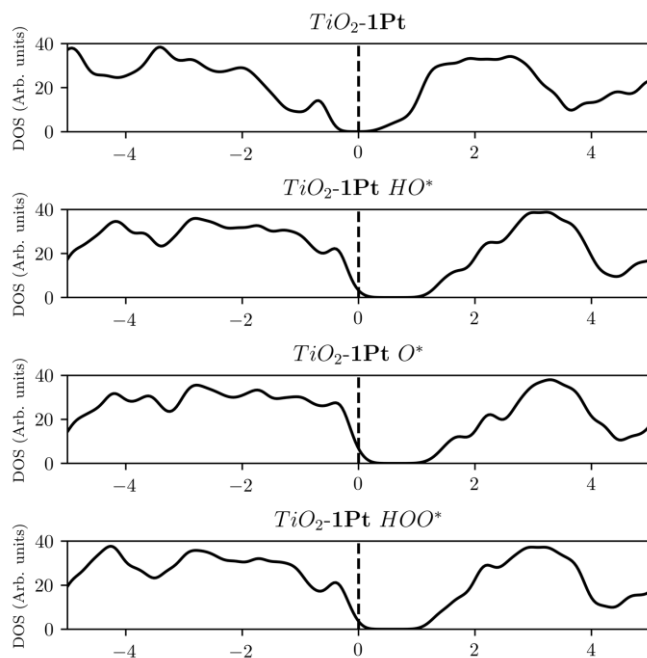
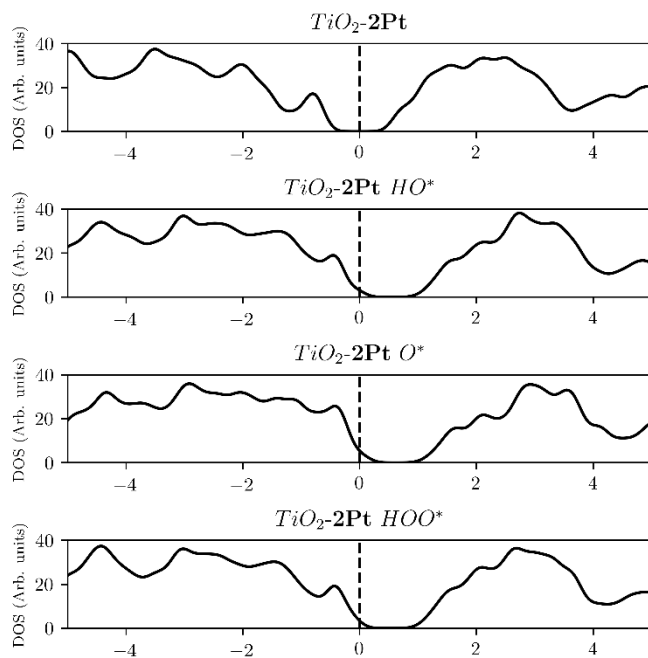
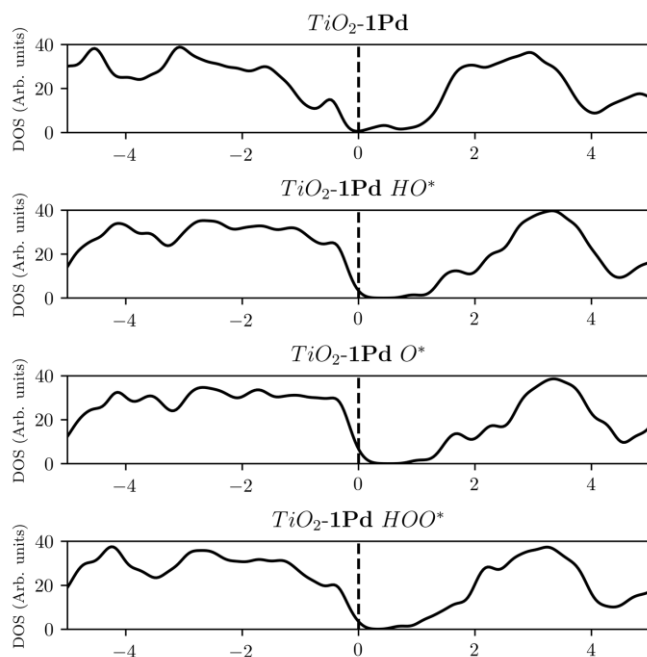
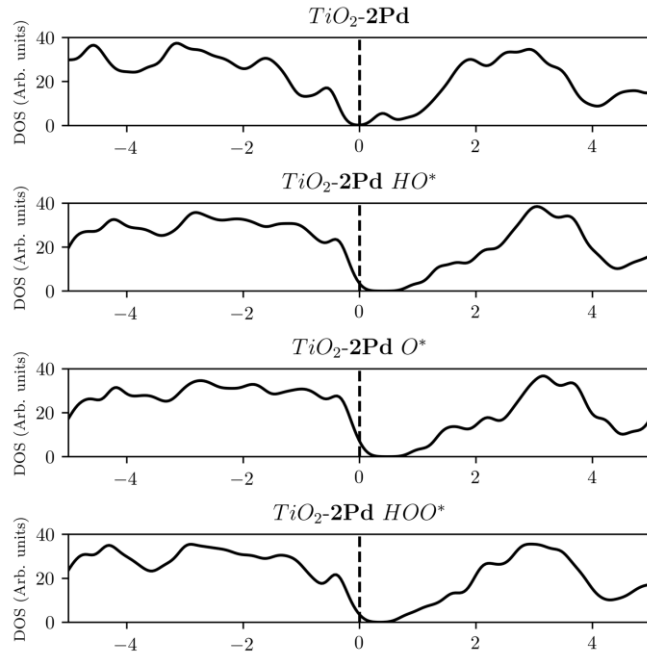
a.**b.**

Fig SI-6: Density of state (DOS) **a.** one dopant (W) in TiO_2 **b.** two dopant (W) in TiO_2

a.**b.****Fig SI-7:** Density of state (DOS) **a.** one dopant (Pt) in TiO_2 **b.** two dopant (Pt) in TiO_2 **a.****b.****Fig SI-8:** Density of state (DOS) **a.** one dopant (Pd) in TiO_2 **b.** two dopant (Pd) in TiO_2

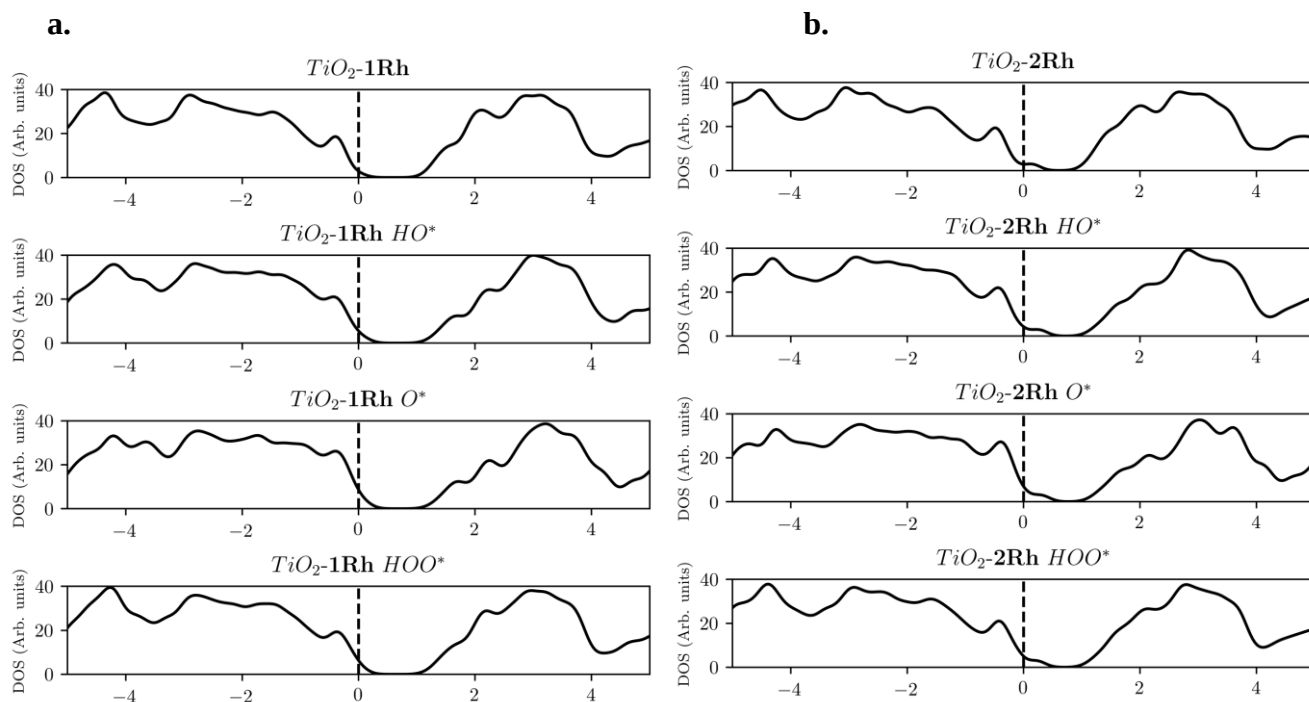


Fig SI-9: Density of state (DOS) **a.** one dopant (Rh) in TiO₂ **b.** two dopant (Rh) in TiO₂

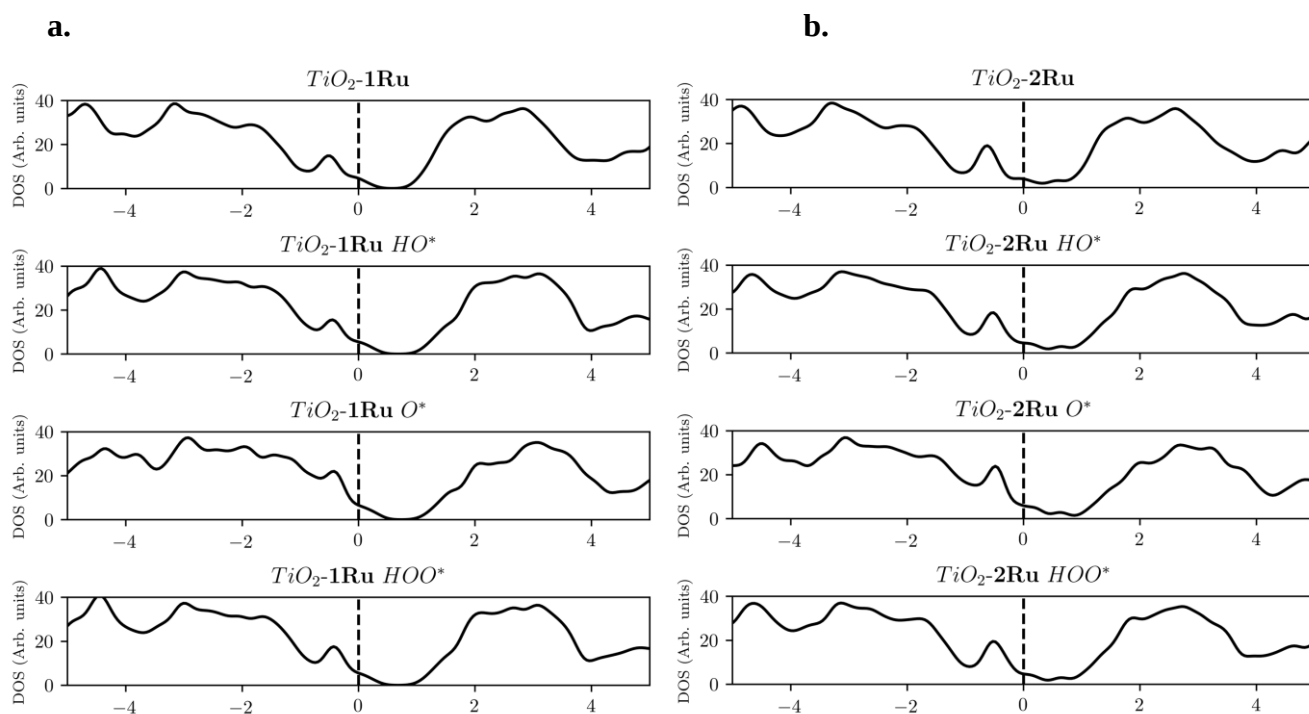
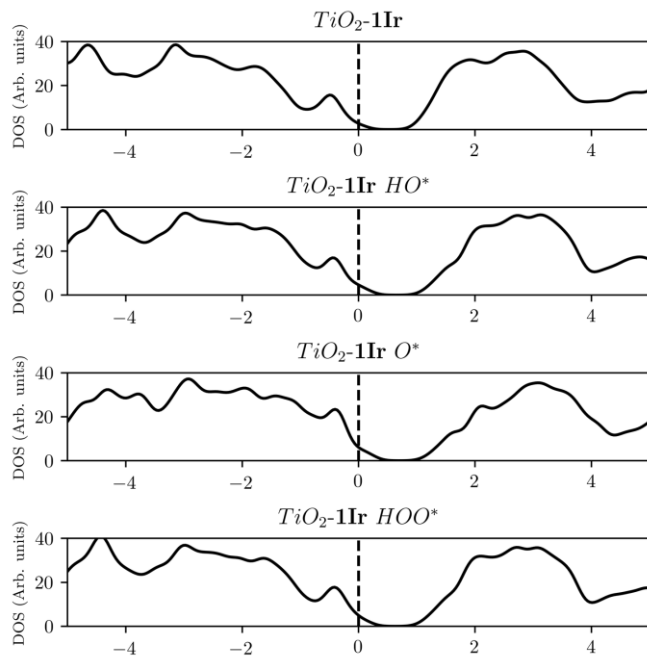
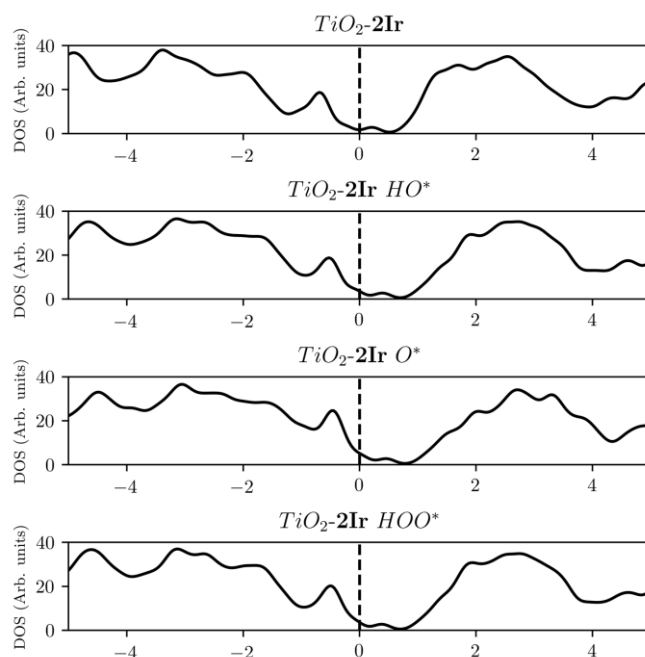
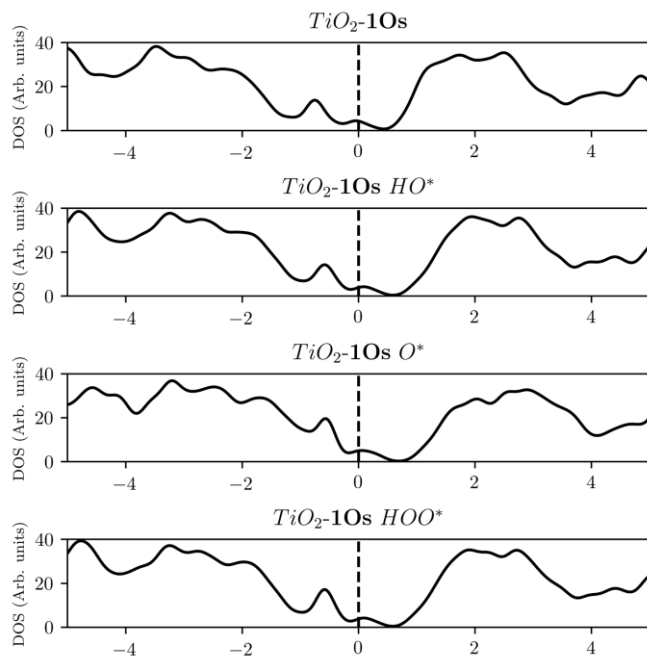
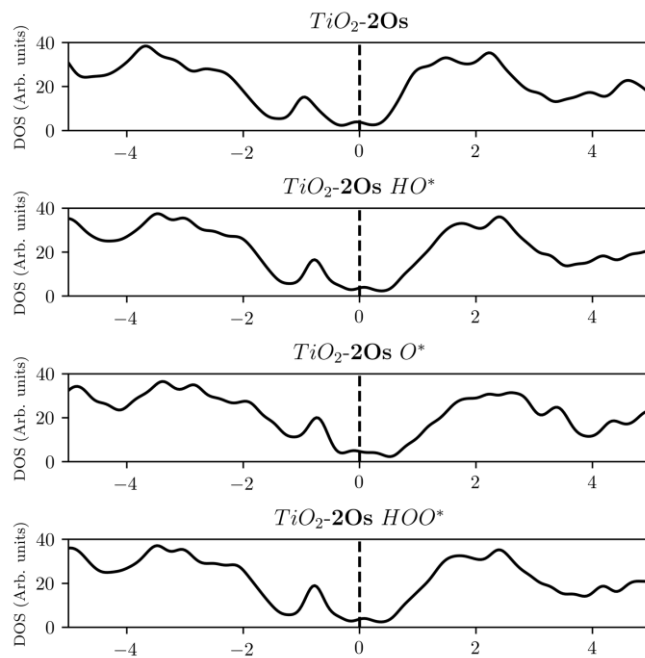
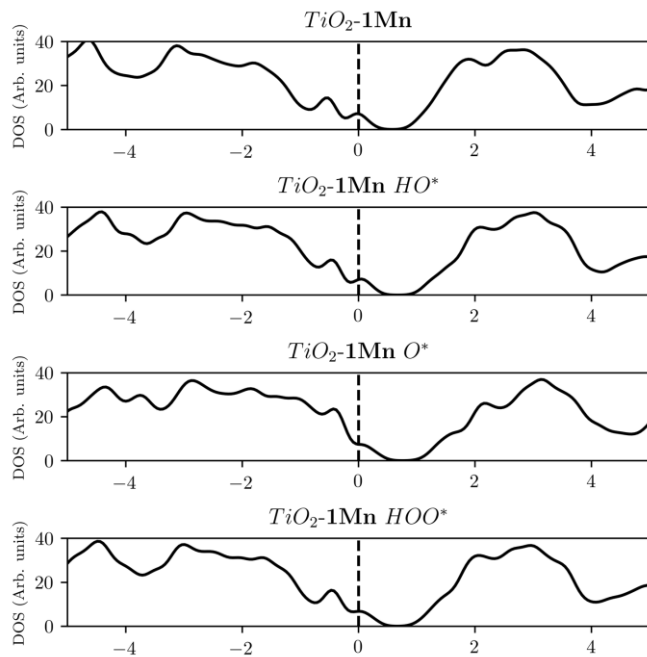
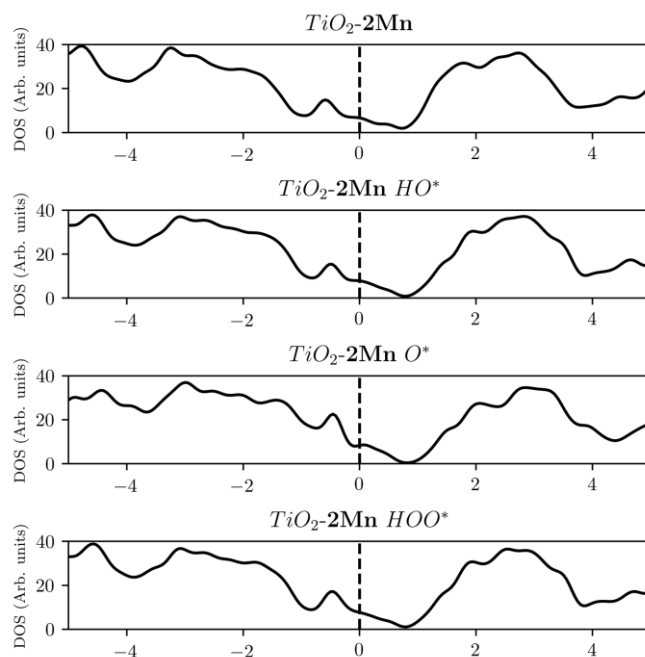
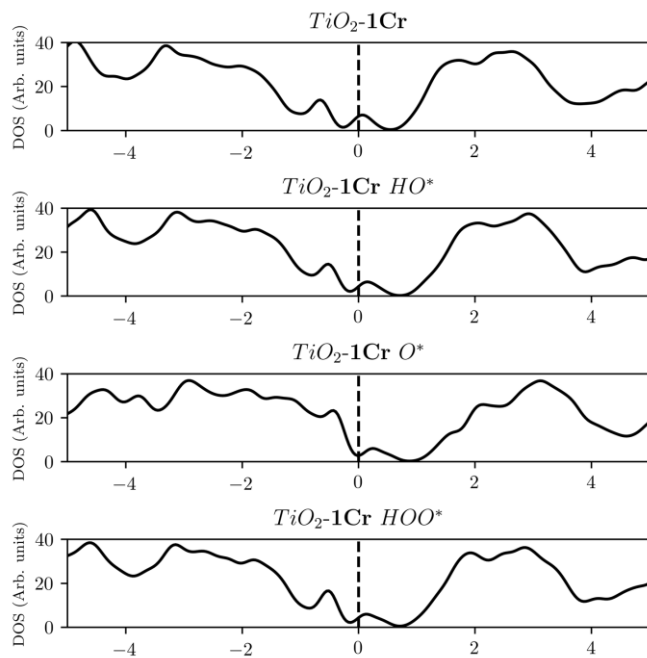
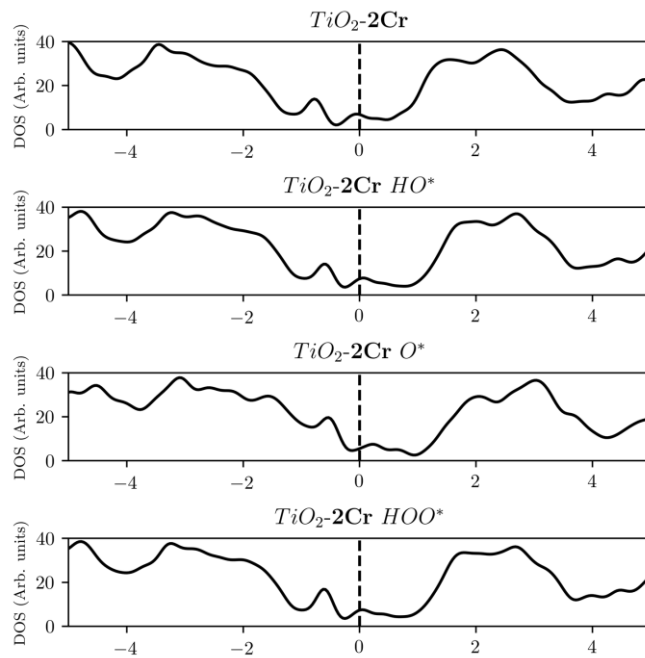
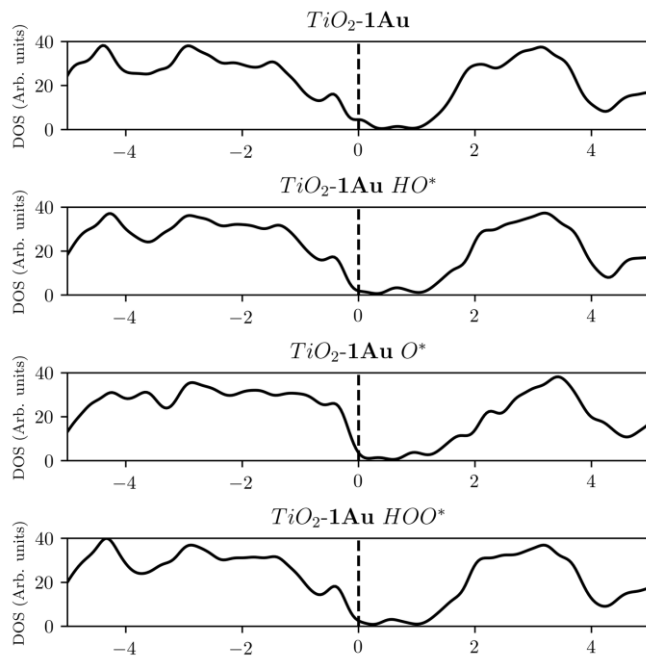
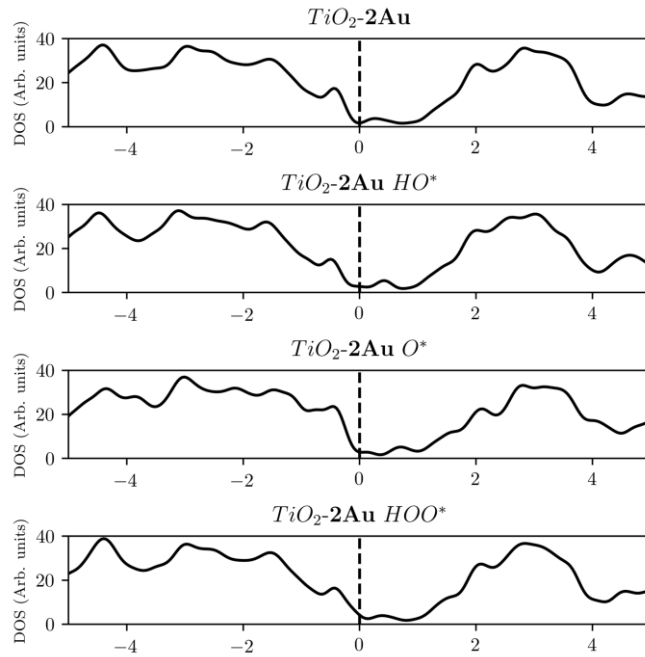
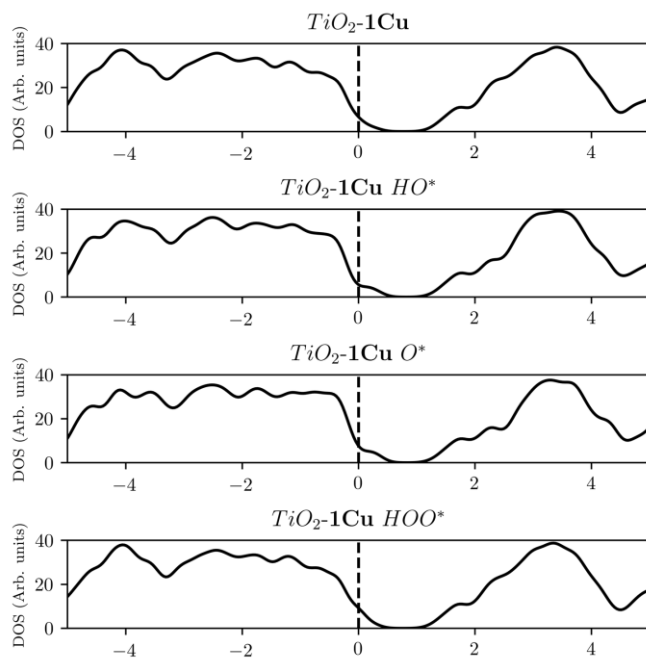
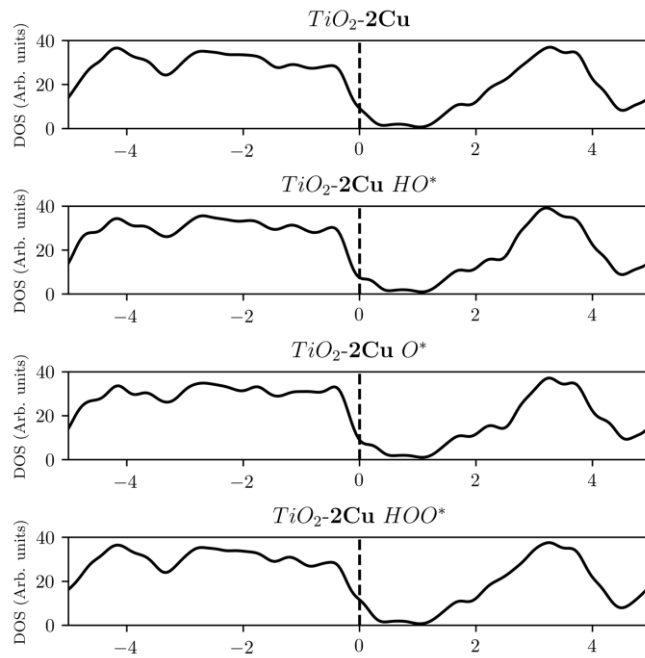


Fig SI-10: Density of state (DOS) **a.** one dopant (Ru) in TiO₂ **b.** two dopant (Ru) in TiO₂

a.**b.****Fig SI-11:** Density of state (DOS) **a.** one dopant (Ir) in TiO_2 **b.** two dopant (Ir) in TiO_2 **a.****b.****b.****Fig SI-12:** Density of state (DOS) **a.** one dopant (Os) in TiO_2 **b.** two dopant (Os) in TiO_2

a.**b.****Fig SI-13:** Density of state (DOS) **a.** one dopant (Mn) in TiO_2 **b.** two dopant (Mn) in TiO_2 **a.****b.****Fig SI-14:** Density of state (DOS) **a.** one dopant (Cr) in TiO_2 **b.** two dopant (Cr) in TiO_2

a.**b.****Fig SI-15:** Density of state (DOS) **a.** one dopant (Au) in TiO_2 **b.** two dopant (Au) in TiO_2 **a.****b.****Fig SI-16:** Density of state (DOS) **a.** one dopant (Cu) in TiO_2 **b.** two dopant (Cu) in TiO_2

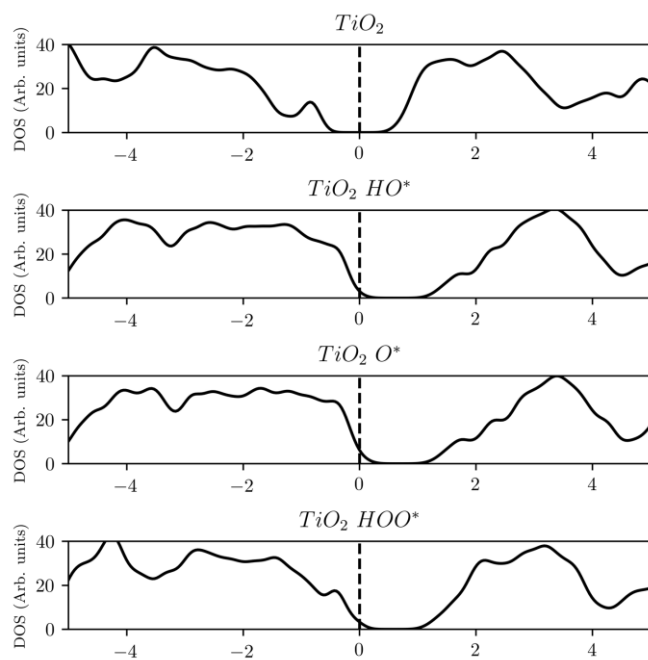


Fig SI-17: Density of state (DOS) for pristine TiO₂

Table SI-4: Adsorption energy of three intermediates (HO*, O*, HOO*) for different dopants (one and two dopants) in TiO₂ structures

	ΔE HO*	ΔE O*	ΔE HOO*
TiO2	1,633077	4,352671	5,046153
2V	0,03641	1,390916	3,637823
2Nb	-0,44958	0,256103	3,084431
2Ta	-0,49677	0,18293	2,926841
2Mo	0,102269	0,992773	3,478655
2W	-0,32801	0,347133	3,225925
2Pt	1,168626	3,750484	4,681546
2Pd	1,473628	4,207047	4,895374
2Rh	1,313538	4,017343	4,848362
2Ru	0,973081	3,160851	4,492736
2Ir	0,912843	3,189322	4,340611
2Os	0,308604	2,158931	3,926341
2Mn	0,991215	3,240015	4,581611
2Cr	0,72088	2,654688	4,187409
2Au	1,197591	3,613883	4,788244
2Cu	1,779615	4,544877	5,136077
2Fe	1,431171	3,975927	4,943121
2Ni	1,478285	4,271062	4,974597
2Co	1,539118	4,309305	5,012429
2Cd	2,394388	5,249781	5,222505
2Re	0,176068	1,41818	3,622421
2Ag	1,556566	4,285838	5,073746
2Hf	1,641076	4,482665	5,027759
2Zn	2,398317	5,291891	5,204763
2Zr	1,622791	4,451426	4,905794
1V	0,113924	2,679152	3,562515
1Nb	-0,40939	2,071601	3,124004
1Ta	-0,43271	2,045836	3,105523
1Mo	0,002361	1,138919	3,412567
1W	-0,46572	0,410023	3,140007
1Pt	1,307547	4,073048	4,718303
1Pd	1,417343	4,239726	4,886083
1Rh	1,553425	4,252662	4,866472
1Ru	1,007363	3,466548	4,472269
1Ir	1,025532	3,535483	4,626116
1Os	0,465855	2,511643	4,06408
1Mn	1,085418	3,605626	4,627375
1Cr	0,742121	3,039522	4,309394
1Au	0,976878	3,589026	4,469841
1Cu	1,808294	4,619463	5,156222

1Fe	1,493925	4,252838	5,069457
1Ni	1,543732	4,357717	5,05759
1Co	1,696888	4,510965	5,130444
1Cd	2,263044	5,168047	5,175158
1Re	-0,08389	1,455379	3,519086
1Ag	1,532312	4,318613	4,936781
1Hf	1,567615	4,379069	4,99412
1Zn	2,403053	5,303774	5,204099
1Zr	1,576064	4,388204	4,975584

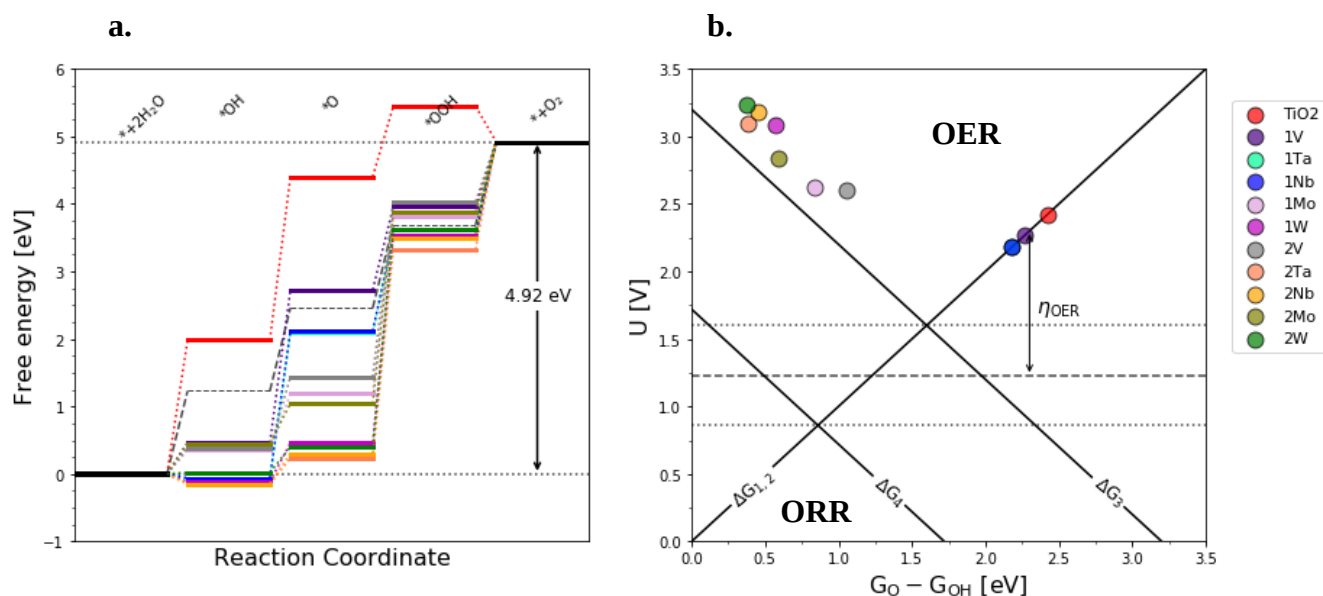
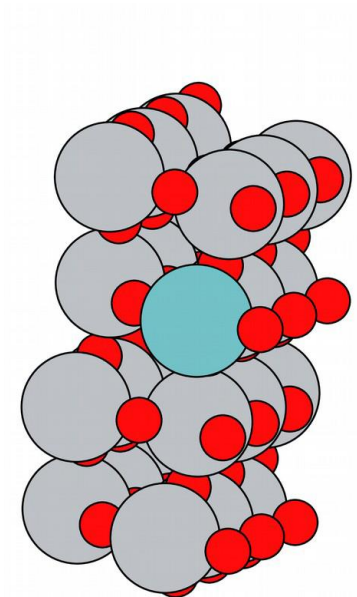


Fig SI-18: a. Gibbs Free Energy of n-type doped TiO2 **b.** Volcano plot for n-type doped TiO2

The calculations containing n-type dopants with sufficient number of electrons to bind O^* without creating the finite size effect (e.g two V or a single W atom in case of TiO_2) are migrating towards the strong side of the volcano. The reason for that, can be seen at the free energy diagram, where the oxygen intermediate is always binding stronger when both electrons of the bond are provided from the dopant.

a.



b.

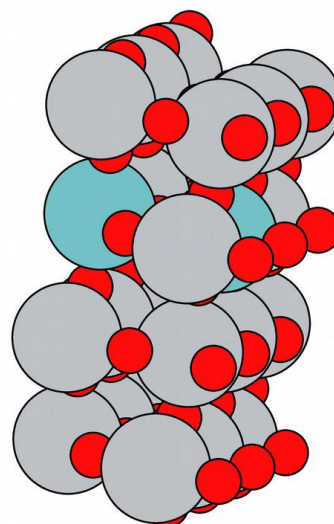
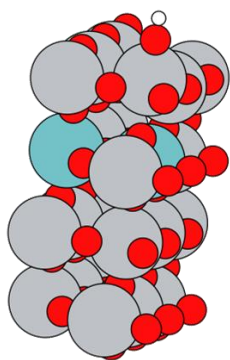
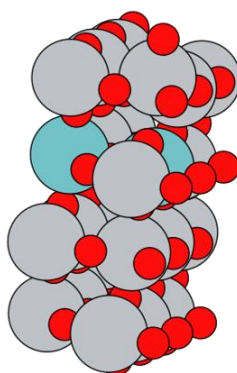


Fig SI-19: a. Illustration of one dopant (Nb) **b.** Illustration of two dopant (Nb)

a.



b.



c.

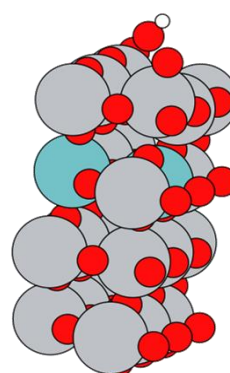


Fig SI-20: Illustration of the intermediates adsorbed on the surfaces. a. Adsorption of HO^* **b.** Adsorption of O^* **c.** Adsorption of HOO^* .

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